MACHINE LEARNING-FACILITATED CROSS-PANEL AUTOMATIC CLASSIFICATION AT BOTH SAMPLE AND CELL LEVEL IN ACUTE MYELOID LEUKEMIA

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INTRODUCTION

Flow cytometry is crucial for diagnosing acute myeloid leukemia (AML), but current manual gating analysis is subjective and ad hoc, without providing needed consistency and accuracy across reagent panels and diagnostic laboratories.

Our previous studies demonstrated that machine learning can predict AML subtypes and detect abnormalities accurately

This study presents two new breakthroughs: we applied the sample-level prediction model across reagent panels and showed success with high classification accuracy and made the machine prediction interpretable by identifying the leukemic cell populations using automated gating algorithms;

MATERIAL AND METHOD

A total of 103 FCS records from the bone marrow samples were used for sample level model (Table 1). Data from RPCCC were acquired using a Beckman Coulter Navios EX and measured with the ClearLLab10C panel (Table 2a); while data from UPMC were acquired using a BD FACSCantoII and measured with a UPMC-developed diagnostic panel (Table 2b).

Table 1. Demographics

Characteristics	All, N (%)	RPCCC	UPMC
Total FC record	**103**	**53**	**50**
Instrument			
FASCanto	50	0	50
Navios EX	53	53	0
Specimen			
Bone Marrow	103	53	50
Flow Diagnosis			
AML	52	27	25
non-neoplastic	51	26	25*

*non-neoplastic pancytopenia
Abbreviation: FC: flow cytometry RPCCC: Roswell Park Comprehensive Cancer Center; AML: acute myeloid leukemia

Table 2. Panel Composition

2a) ClearLLab10C

Tube	FITC	PE	ECD	PC5.5	PC7	APC	APC-A700	APC-A750	Pacific Blue	Krome Orange
B	Kappa	Lamda	CD10	CD5	CD200	CD34	CD38	CD20	CD19	CD45
T	TCRγδ	CD4	CD2	CD56	CD5	CD34	CD7	CD8	CD3	CD45
M1	CD16	CD7	CD10	CD13	CD64	CD34	CD14	HLA-DR	CD11b	CD45
M2	CD15	CD123	CD117	CD13	CD33	CD34	CD38	HLA-DR	CD19	CD45

2b) UPMC developed diagnostic panel

Tube No.	FITC	PE	PerCP-Cy5-5	PE-Cy7	APC	APC-H7	V450	V500
1	CD36	CD123	CD64	CD33	CD34	CD14	HLA-DR	CD45
2	CD15	CD56	CD117	CD13	CD11b	CD16	HLA-DR	CD45
3	Kappa	Lambda	CD5	CD19	CD10	CD38	CD20	CD45
4	CD16&57	CD7	CD4	CD3	CD56	CD8	CD2	CD45
5	CD7	CD13&33	CD19	-	CD56	-	-	-

Sample Level Classification and Feature Selection

We applied a previously published Gaussian Mixture Model-Support Vector Machine (GMM-SVM) based approach to train a cross panel sample-level classification model for AML diagnosis (Figure 1)(1-3). A three-fold cross-validation was conducted to assess the model performance, using sensitivity, specificity, classification accuracy, and area under the receiver operating characteristic curve (AUC). Cross-channel interaction was evaluated via the feature selection experiments. The parameter that permitted the highest ACC was then paired with the individual remaining parameters to determine which pair provided the highest ACC. For each subsequent step, we added the remaining parameters individually to determine the combination with the next-highest ACC.

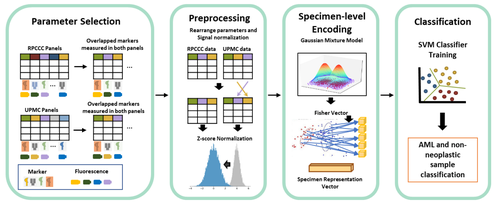


Figure 1. Schematic of Sample Level Classification Model Development
We utilized flow cytometry listmode data (FCS3.1), treating each light scatter property and fluorescent marker as a distinct parameter. Data preprocessing involved parameter alignment and z-score normalization. A Gaussian mixture model (GMM) was employed to capture high-dimensional cellular distribution, followed by Fisher vectorization to embed phenotype characteristics into a high-dimensional phenotype representation vector at the specimen level. These vectors, along with their corresponding diagnoses, were input into a support vector machine (SVM) for AML versus non-neoplastic classification.

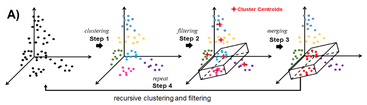
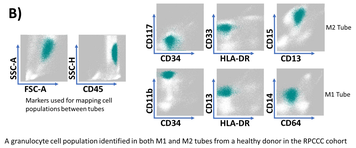


Figure 2. A) Steps in DAFi identification of cell populations; B) Combination of FLOCK clustering and DAFi to identify the same cell populations and map them across two tubes (ClearLLab M1 and M2)



RESULTS

Specimen-level Classification and Visualization

The AML detection model, trained with all parameters on the RPCCC dataset ClearLLab10C panel, achieved an AUC of 100% (Model A). However, when only using overlapping parameters between ClearLLab10C and UPMC panels, the model's AUC marginally decreased to 99.86% (Model B) and 99.52% (Model C) on RPCCC and UPMC datasets, respectively. The cross-panel classifier (Model D), trained on a combined RPCCC and UPMC dataset using 25 overlapping parameters, achieved an AUC of 99.06%. Additionally, by adjusting the model's threshold, the false negative rate (FNR) could be reduced to 0% in Model D, while maintaining a positive predictive value of 92.86% (Table 3).

The visualization of samples from Model D for both RPCCC and UPMC is shown in Figure 3.

Table 3. Sample Level Classification Model Performance

Model version	A	B	C	D
Parameters	all markers measured in ClearLLab10C	25 common markers measured in both ClearLLab10C and UPMC-developed diagnostic panels		
Dataset	RPCCC only	RPCCC only	UPMC only	**RPCCC+UPMC**
Sample size	53	53	50	**103**
Threshold	0.542	0.27	0.27	**0.27**
ACC	100.00%	98.10%	96.00%	**96.10%**
AUC	100.00%	99.86%	99.52%	**99.06%**
sensitivity	100.00%	100.00%	100.00%	**100.00%**
specificity	100.00%	92.31%	84.00%	**92.16%**
PPV	100.00%	93.10%	86.20%	**92.86%**
NPV	100.00%	100.00%	100.00%	**100.00%**
FPR	0.00%	7.69%	16.00%	**7.84%**
FNR	0.00%	0.00%	0.00%	**0.00%**

Abbreviation: RPCCC - Roswell Park Comprehensive Cancer Center; AML - Acute Myeloid Leukemia. ACC: accuracy, defined by the concordance of model prediction and the diagnosis of the sample, AUC: area under the ROC curve
25 markers measured in both ClearLLab10C and UPMC panel: CD2, CD3, CD4, CD5, CD7, CD8, CD10, CD11b, CD13, CD14, CD15, CD16, CD19, CD20, CD33, CD34, CD38, CD45, CD56, CD64, CD117, CD123, HLA-DR, Kappa, Lamda
*Note: In the context of this study, "normal" samples from RPCCC included individuals with lymphoma but no bone marrow involvement, bone marrow donors, residual specimens from hip replacement surgery. Similarly, "normal" samples from UPMC referred to patients with cytopenia.

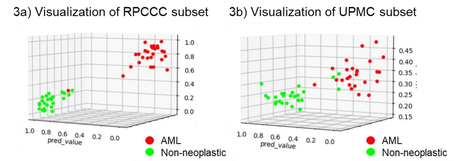


Figure 3. Visualization of AML vs. Non-Neoplastic Classification at Sample Level.
Dimension reduction was applied to each sample-level phenotype vector and the decision score from the SVM classifier depicted in Figure 1. Each sample is represented as a dot on a 3D plot, with colors indicating the actual diagnosis.
3a) Visualization of RPCCC dataset in Model D, 3b) Visualization of UPMC dataset in Model D.

Feature Selection Analysis

The GMM-SVM approach was applied to individual parameters in Model D to assess the relative importance of each parameter to the model's performance. It was found that a model developed using a combination of parameters - CD117, CD123, CD15, CD34, FSC-A, FSC-H, SSC-A, and SSC-H - achieved an equivalent performance with an accuracy (ACC) of 97.01% (Figure 4). This is compared to the 96.01% ACC of Model D (Table 4).

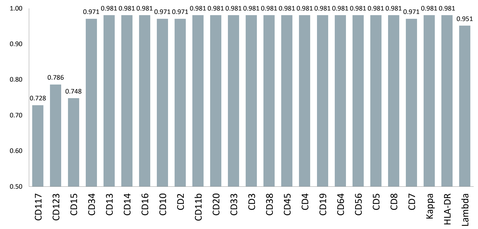


Figure 4. Cross-Channel Interaction Assessment Through Feature Selection Analysis
A combination of FSC-A, FSC-H, SSC-A, and SSC-H was used as a baseline. The parameter that yielded the highest accuracy (ACC) was then paired with each remaining parameter to determine the pair that provided the highest ACC. In each subsequent step, we individually added the remaining parameters to find the combination yielding the next highest ACC.

Cluster Identification and Visualization

The UMAP plots, created by pooling non-neoplastic and AML samples, exhibited distinct patterns for different AML subtypes (M1 to M5) and a genetic mutation (e.g., AML IDH1), which were distinct from healthy control samples (Figure 5).

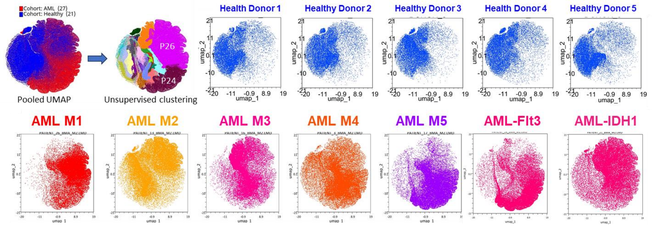


Figure 5. UMAP Visualization of AML samples in RPCCC cohort (27 AML + 21 Healthy). The same pattern of distribution of cell populations was observed across healthy samples, while each subtype of AML has a unique pattern. The three distinctive UMAP regions correspond to immature blasts (top right on the UMAP), healthy cells (middle left on UMAP), and leukemic monocytes (bottom right).

Leukemic Cells Identification and Characterization

Using the automated gating analysis, cell populations identified as different between AML and control samples were characterized and visualized in 2D scatter plots for straightforward interpretation and assessment based on their immunophenotype.

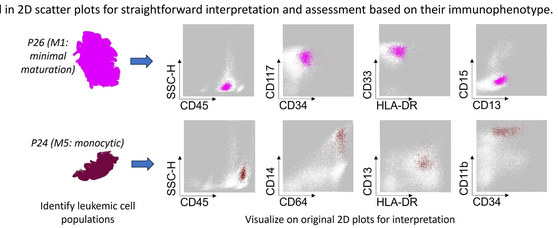


Figure 6. Validation and interpretation of the ML-identified cell populations . Cell population were identified using FLOCK and DAFi, before being visualized on the UMAP and in the original 2D-by-2D dot plots. Upper row: immature blasts seen in the top right region of the UMAP from an M1 subtype patient (P26) were visualized using markers in the ClearLLab M2 tube including CD34, CD117, CD33, and CD13; Bottom row: monocytic leukemic cells seen in the bottom right region of the UMAP from an M5 subtype patient (P24) were visualized using markers in the ClearLLab M1 tube including CD14, CD64, CD34, and CD11b).

Clustering Analysis and Visualization

Using UMAP and FlowSOM, high-parameter flow cytometry data from 48 samples was visualized, with equal cell numbers (50,000 cells/sample) combined for UMAP creation. FlowSOM was used to identify 40 meta-clusters in the 12-dimensional space. The cluster information was used to color-coded cells on the UMAP. Individual sample cells were plotted on the same UMAP.

Identification and Characterization of Leukemic Cells

FlowSOM was utilized for identifying cohort-level biomarkers such as leukemic blasts, as visualized on the UMAP. FCS files underwent FCSTrans transformation before the FLOCK clustering process, which identified clusters of cells in individual samples (4,5). Based on the coordinates of the cell clusters, DAFi was applied to identify the leukemic cell populations following a sequential 2D gating strategy (6). The DAFi-identified leukemic cell populations were then visualized in 2D-by-2D plots using the same marker pairs as in the manual gating strategy, providing phenotypic annotations and interpretation for each cell population.

CONCLUSIONS

- **The GMM-SVM based approach successfully developed a sample level classification model with high performance in distinguishing AML from non-neoplastic samples.**
- **This approach allows flexibility in using data from different instrument models and reagent panels.**
- **A cross-panel AML classification model with equivalent performance can be developed using only optical parameters in conjunction with CD117, CD123, CD15, and CD34, as opposed to using 25 parameters.**
- **The use of UMAP and FlowSOM aids in visualizing differences among AML subtypes across 40 clusters and identifying clusters specific to particular subtypes.**
- **Combination of FLOCK clustering with DAFi identification of cell populations was able to identify leukemic cell populations of phenotypic heterogeneity within individual samples**
- **Further analysis is required to validate these findings in a broader context.**

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